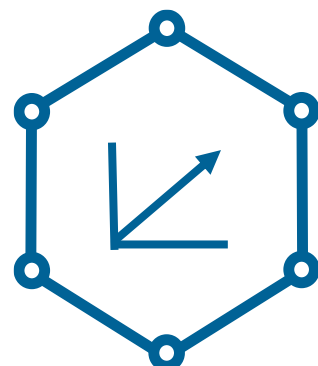


Knowledge Graphs

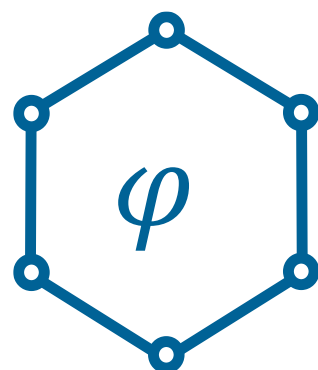


Emanuel Sallinger



KG Embeddings

Widely-applied, large family of ML models



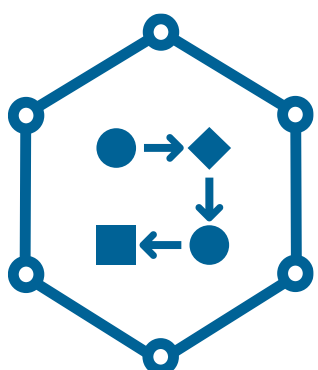
Logical Knowledge in KGs

Highly expressive, diverse family of logical models.



Graph Neural Networks

Using the KG structure as a neural network.



Data Models

Overview of data models in different communities



Graph Neural Networks



Message Passing Neural Networks



GraphSAGE



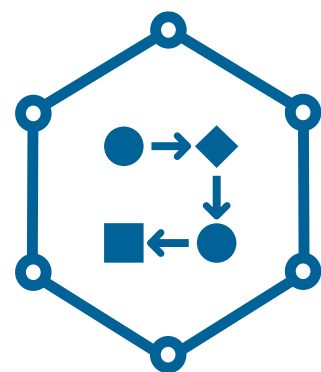
Graph Convolutional Networks



Applications



Theory to Practice



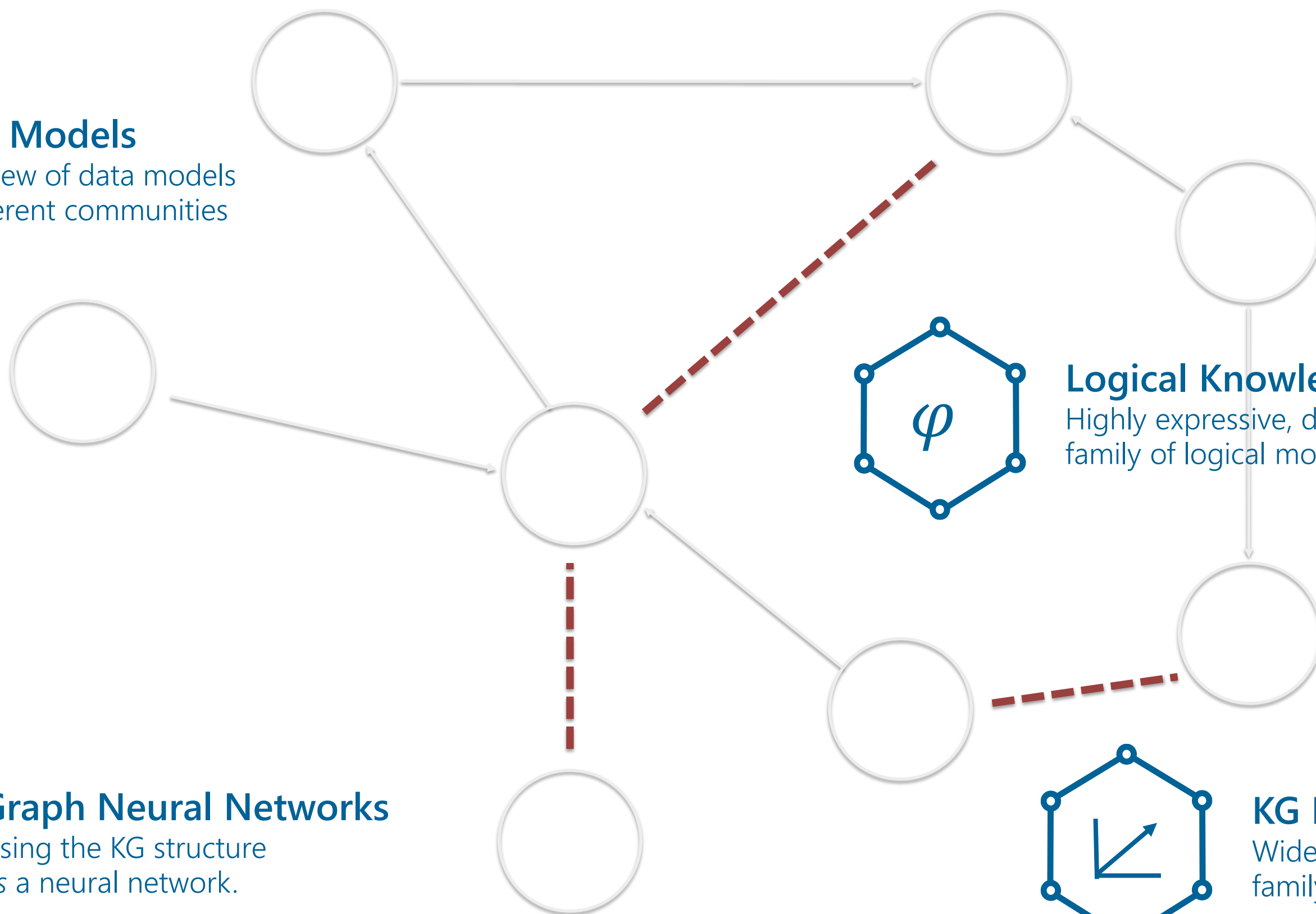
Data Models

Overview of data models in different communities



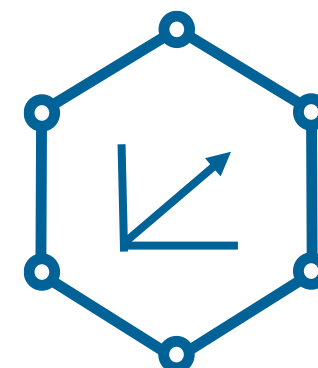
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KG Embeddings

Widely-applied, large family of ML models



$$KG = (D, K; g, l, r)$$

Obtaining
Learning

Applying
Reasoning





Knowledge Graph Embeddings: **Latent Knowledge**

$$KG = (D, K; g, l, r)$$

$$\mathbb{K} = \mathbb{e}$$

$$l: \mathbb{D} \rightarrow \mathbb{e}$$

$$r_{\mathbb{e}}: \mathbb{D} \rightarrow \mathbb{D}$$



Logical Knowledge in KGs:

Explicit Knowledge

$$KG = (D, K; g, l, r)$$

$$K = \Sigma$$

$$l: \mathbb{D} \rightarrow \Sigma$$

$$r_{\Sigma}: \mathbb{D} \rightarrow \mathbb{D}$$



Graph Neural Networks:

Latent Knowledge

$$KG = (D, K; g, l, r)$$

$$\mathbb{K} = \mathbb{W} \text{ (w.r.t. } F)$$

$$l: \mathbb{D} \rightarrow \mathbb{W}$$

$$r_{\mathbb{W}}: \mathbb{D} \rightarrow \mathbb{D}$$



Graph Neural Networks



Message Passing Neural Networks



GraphSAGE



Graph Convolutional Networks



Applications



Theory to Practice



The aim of this is course is to gain a deep understanding of Knowledge Graphs divided into three blocks. An overarching aim of the course is to understand the connections between Knowledge Graphs, Artificial Intelligence, Machine Learning and Data Science.

Representations

Logical and ML-based representations for KGs.

Read more details



KG Embeddings

Widely-applied, large family of ML models.



Logical Knowledge in KGs

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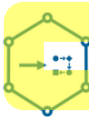
Architectures

The big picture of building IT architectures for KGs.



Scalable Reasoning

Making use of the knowledge in the KG.



KG Creation

How to create a KG from heterogeneous data?



KG Evolution

How to update, correct and complete a KG?

Systems

Systems to bring KGs into practice.

Read more details

Applications

Real-world applications of KGs.

Read more details



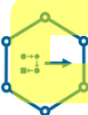
Real-World Applications

Overview of diverse applications.



Financial KGs

Concrete applications in finance and economics.



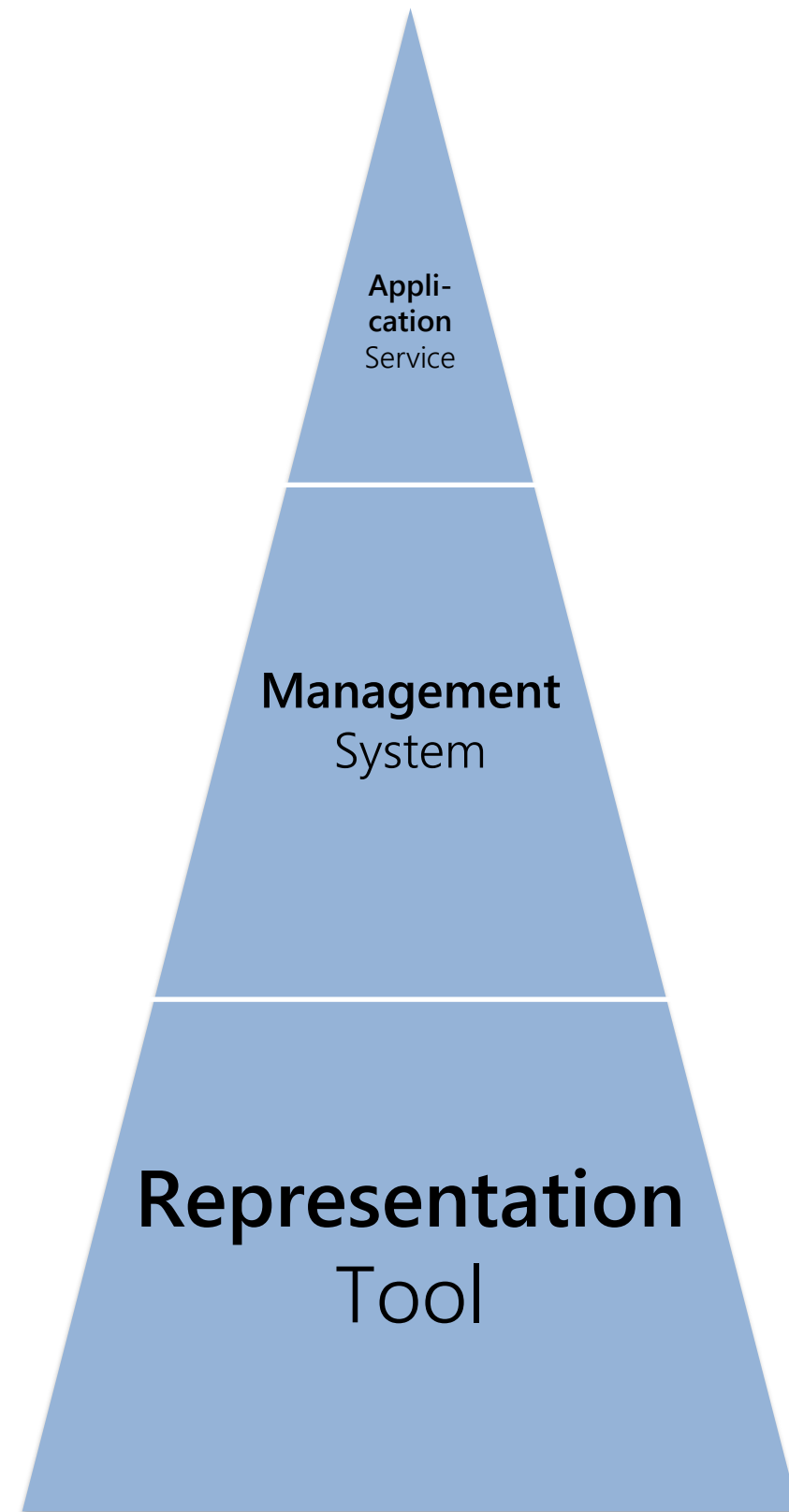
Services

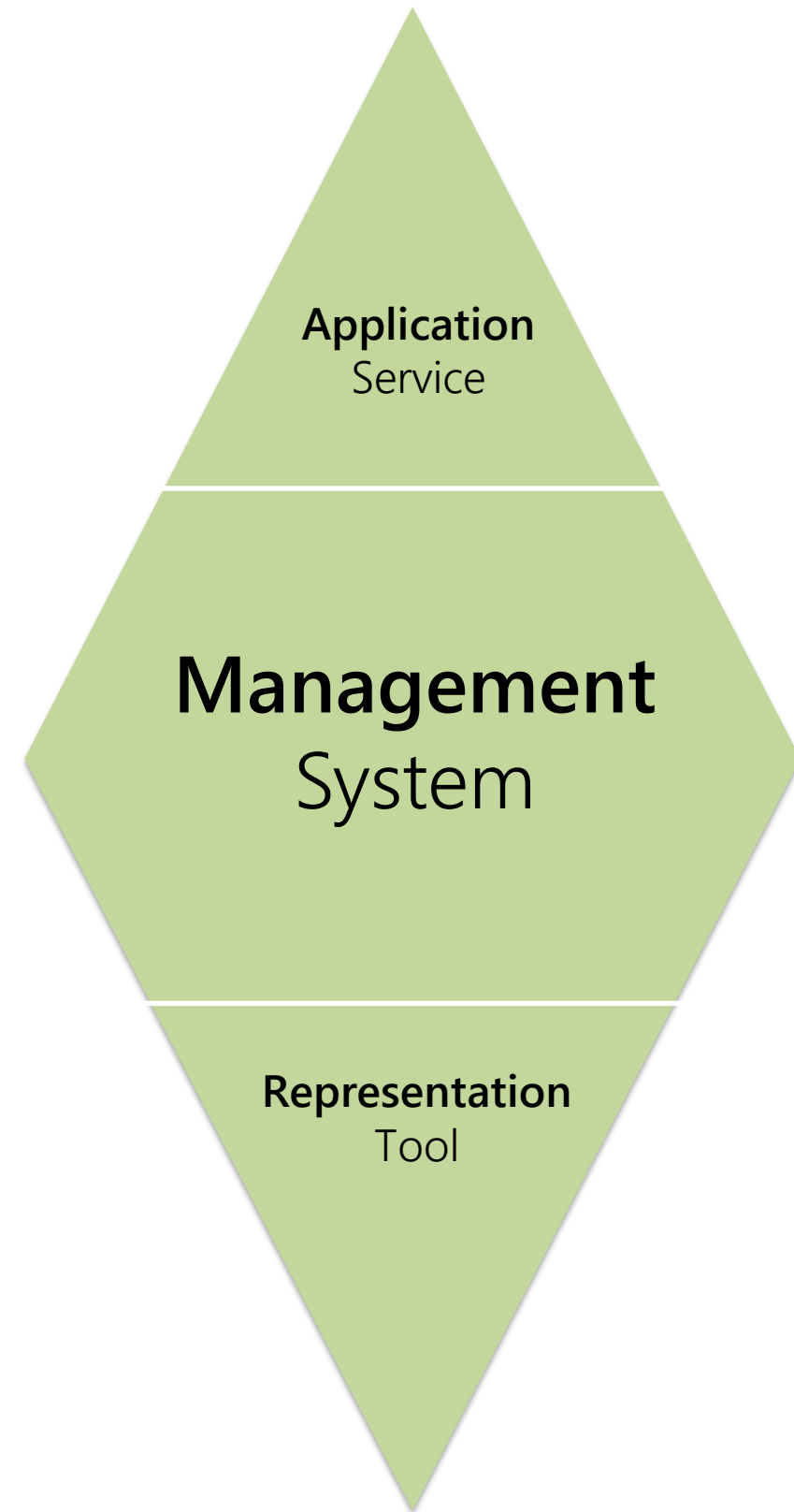
Which service to provide based on KGs?

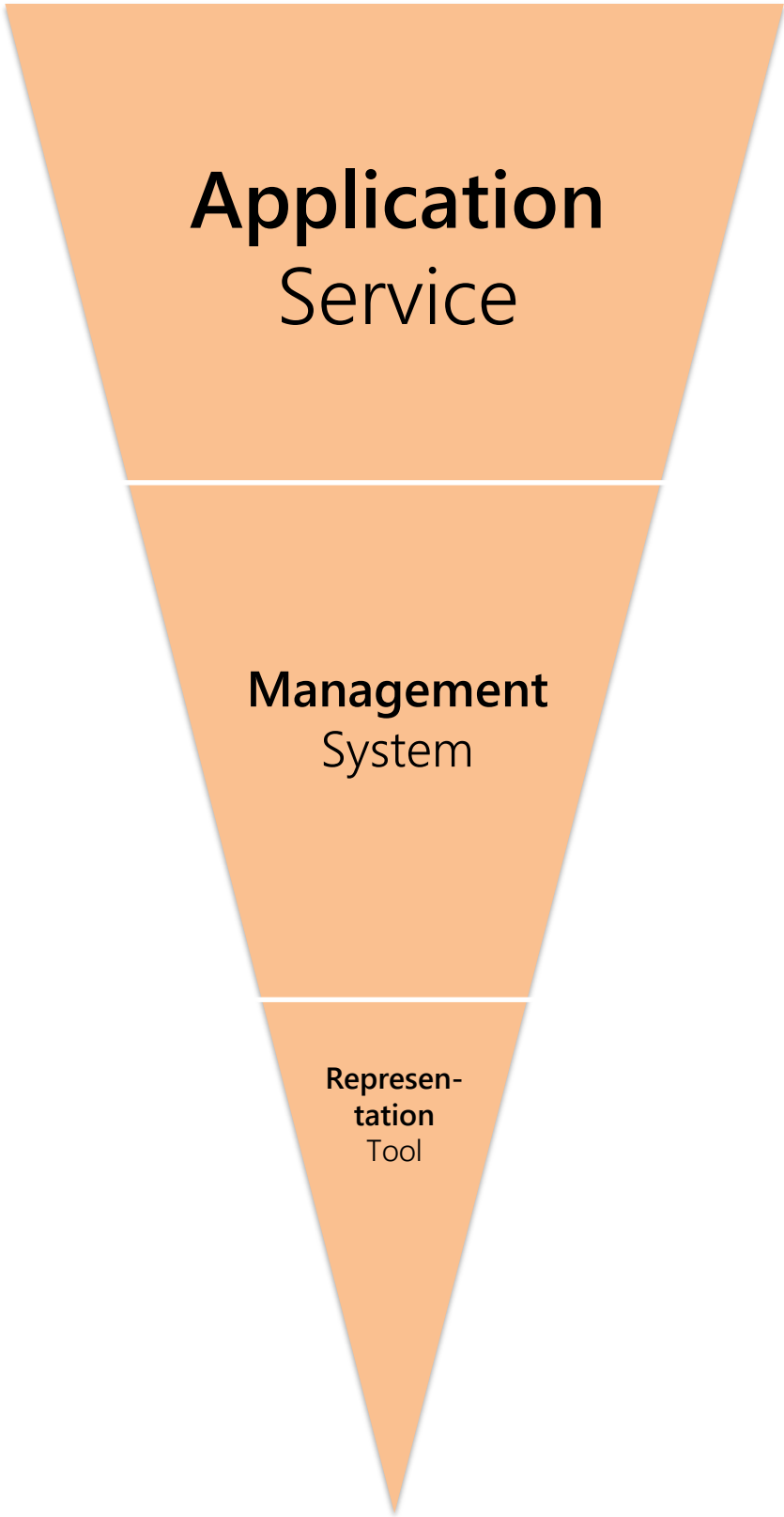


Connections

... between KGs, AI, ML and Data Science.









Graph Neural Networks



Message Passing Neural Networks



GraphSAGE



Graph Convolutional Networks



Applications



Theory to Practice